INTERNATIONAL MONETARY FUND

LAO PEOPLE'S DEMOCRATIC REPUBLIC

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STAFF REPORT FOR THE 2023 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staff of the International Monetary Fund and the International Development Association (IDA)[1]

Risk of external debt distress	In debt distress
Overall risk of debt distress	In debt distress
Granularity in the risk rating	Unsustainable
Application of judgement	Yes. Given on-going negotiations about debt service deferral and significant and sustained breaches of debt thresholds

The Lao P.D.R. is assessed to be in external and overall debt distress under the Low-Income Countries Debt Sustainability Framework (LIC-DSF). Compared to the 2019 Debt Sustainability Analysis[2], the country's risk rating and debt sustainability have deteriorated due to an adverse macroeconomic environment (through a strong exchange rate depreciation), an expansion of debt coverage to include guaranteed debt, expenditure arrears, and the issuance of domestic debt to recapitalize state-owned banks.[3] PPG and external debt are projected to decline in the baseline scenario, reflecting small primary surpluses and economic growth above interest rates, but the present value of PPG debt-to-GDP ratio remains very high and above the indicative threshold for two decades. Most of the PPG external and public debt indicators for both solvency and liquidity have persistent breaches of their indicative thresholds. The on-going negotiations about debt

[1] This DSA has been prepared jointly by the IMF and World Bank, following the 2018 Guidance Note on the Bank-Fund Debt Sustainability Framework for Low-Income Countries.

[2] Country report No.19/267, August 2019.

[3] The Lao P.D.R. Composite Indicator of 2.15 indicates a weak debt-carrying capacity, based on the October 2022 IMF's World Economic Outlook (WEO) and the 2021 World Bank's Country Policy and Institutional Assessment (CPIA).

service deferral indicate debt distress and the significant and sustained breaches of debt thresholds in the DSA also indicate a high probability of a future distress event.This result, along with the evidence of major economic vulnerabilities (including significant financing needs, low foreign exchange liquidity and reserves, limited external market access since 2020, limited potential for financing from domestic capital markets, increasing rollover risks, potential for additional exchange rate depreciation pressures, possible accumulation of additional domestic arrears, and substantial contingent liabilities), warrants, through the application of judgement, a rating of "in debt distress" for both external and public debt and an assessment of unsustainable public debt. Other DSA results show that external debt is most vulnerable to shocks to exports and a depreciation of the currency, while public debt is most sensitive to the contingent liabilities shock. Adjustment options are limited, as the primary surpluses required to reduce debt indicators below risk thresholds within a few years are not politically feasible or socially desirable.

PUBLIC DEBT COVERAGE

1. The debt coverage in this Debt Sustainability Analysis (DSA) comprises debt owed by the central government and government guarantees, including to state-owned enterprises (SOEs) (Text Table 1). Publicly guaranteed SOE external debt included in the DSA baseline is on-lent or guaranteed by the central government. There is no outstanding debt to the IMF. External debt is defined based on the currency criterion in this analysis. The debt stock includes a swap line with the People's Bank of China (PBoC) in 2020, given that the swap appears to be for balance of payments support to help shore up gross reserves. The stock also includes past deferred interest payments in 2021 and 2022, and expenditure arrears. There have been ongoing efforts by the authorities to enhance debt transparency: the government published an annual debt statistics bulletin for the first time in 2021 that includes publicly guaranteed external debt and domestic debt contracted by the central government, and the following bulletin included more detailed information.[4] Although the data in the debt bulletin provides information by creditor, information on debt owed by local governments and non-guaranteed SOE debt is lacking. Additionally, no information is available on the debt bulletin regarding borrowing of the central bank on behalf of the government. Therefore, the DSA coverage does not include these subsectors, and the contingent liability

Text Table 1. Coverage of Public Debt

	Subsectors of the public sector	Sub-sectors covered
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	
8	Non-guaranteed SOE debt	

[4] In addition to the information reported in the 2020 bulletin (published in 2021), the 2021 debt bulletin (issued in 2022) included the composition of on-lending and guaranteed debt by SOE, as well as expenditure arrears by type. The International Development Association (IDA), through its Sustainable Development Finance Policy (SDFP) Performance and Policy Actions (PPAs), has supported reforms to enhance debt transparency (publication of an annual public debt bulletin), debt management (compliance with a non-concessional borrowing ceiling), and fiscal sustainability (approval of a decree on SOEs).

stress test is used with tailored settings under the DSF to investigate potential effects of the uncovered public debts.

2. A customized contingent liability stress test is used to analyze the impact of additional public debt coming from contingent liabilities from PPPs and SOEs, additional occurrence of government arrears in the future, and recapitalization needs from the banking sector (Text Table 2).[5] Given risks of SOEs default, the contingent liability stress test assumes an additional shock related to SOEs by 8.3 percent of GDP in the second year of the projection. The parameter includes; (i) US$ 890 million (6.3 percent of GDP) coming from potential need to cover guaranteed state enterprise debt, and (ii) the default parameter value (2 percent of GDP) to capture non-guaranteed SOEs, which is not included in the debt coverage.[6] The shock related to PPPs is set to the default value of 35 percent of the PPP capital stock (equivalent to 26.5 percent of GDP) to capture contingent liabilities from the private sector share of PPPs.[7] A shock covering 0.5 percent of GDP has been added to analyze the effect of the discovery of potential domestic arrears. The test assumes a financial market shock of 11.5 percent of GDP, which includes LAK 4 trillion (1.5 percent of GDP) of potential commercial bank recapitalization and twice the default value (10 percent) given the financial sector remains weak.

Text Table 2. Calibration of Shock Scenarios

1	The country's coverage of public debt	The central government, government-guaranteed debt		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.5	accounts for potential discovery of arrears
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	8.3	captures non-guaranteed domestic SOEs debt and restructuring of SOEs.
4	PPP	35 percent of PPP stock	26.47	captures the contingent liabilities of the private sector share of PPPs
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	11.5	captures bank recapitalization needs and risks from a sensitivity of the financial market.
	Total (2+3+4+5) (in percent of GDP)		46.8	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

DEBT DEVELOPMENTS

3. The authorities have established a Public Debt Management Committee led by the Deputy Prime Minister. All borrowing (by central and provincial governments, SOEs, Public-Private Partnerships (PPPs)) must be approved by the National Assembly and the Ministry of Finance negotiates all loans. To improve the legal and regulatory framework, the government has implemented a decree which mandates SOEs to report debt data to the Ministry of Finance and to consult the ministry before a significant borrowing.

[5] The DSA assumes that the additional financing needs due to the stress test shock will be covered mainly by domestic financing, given the limited access to external borrowing in the next few years.

[6] For guaranteed SOE debt, staff assumes an additional public debt increase equivalent to one-sixth of its outstanding debt as of end-2021.

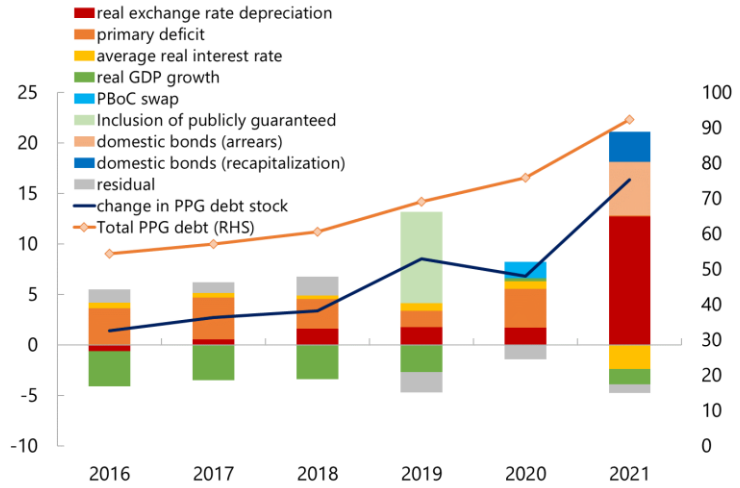
[7] Data on PPP capital stock are from the World Bank's Private Participation in Infrastructure Database and is available at https://ppi.worldbank.org/.

4. The Lao P.D.R.'s total public and publicly guaranteed (PPG) debt is estimated to be 92.4 percent of GDP as of end-2021.

- Most of the PPG debt is external, representing 86.1 percent of the outstanding public debt or 79.5 percent of GDP (Text Table 3). Publicly guaranteed debt is included in the external PPG debt stock for 2019 and beyond, and the effect of the inclusion in 2019 accounts for a large portion of debt increase which is not explained by automatic debt dynamics (Text Figure 1).[8] In 2021, guaranteed debt represents 15.8 percent of total PPG external debt stock. The DSA also treats as external PPG debt a swap line with the People's Bank of China (PBoC) of US$ 300 million in 2020, with a full repayment expected in 2023. Lao P.D.R. relies mainly on bilateral borrowing, in which China is the largest creditor. As a result, China has the largest share—nearly half—of external PPG debt (Text Figure 2). Multilateral creditors–mainly the International Development Association (IDA) and the Asian Development Bank (ADB)—account for 12.6 percent of external PPG debt. The remaining of external PPG debt is from sovereign bonds and commercial banks, which account for 7.9 percent and 7.5 percent of the external PPG debt stock, respectively. In terms of public external debt, 57.4 percent is on concessional or semi-concessional terms. The authorities' Debt Bulletin indicates that debt servicing to China was suspended: principal repayments in 2020 of about US$ 220 million, and of both principal and interest payments in 2021 and 2022 of about US$ 450 million and US$ 610 million, respectively.[9]

Text Figure 1. Change in PPG debt contributions

(Percentage points of GDP)

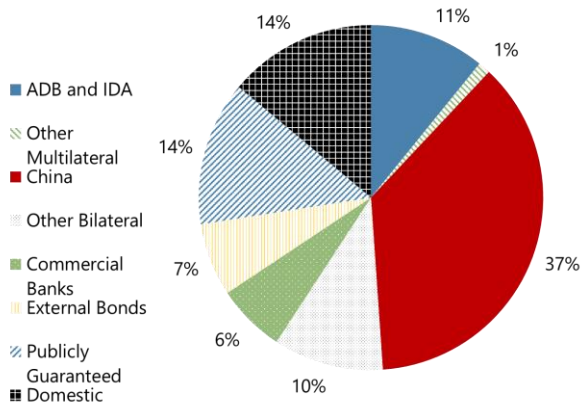


Sources: National authorities; and IMF staff calculations and projections.

[8] Publicly guaranteed debt in 2019 accounted for 9.2 percent of GDP.

[9] The Lao P.D.R. did not participate in the Debt Service Suspension Initiative (DSSI), with the authorities preferring direct discussions with key bilateral creditors.

Text Figure 2. Stock of Public Debt as of end-2021
(In percent of total PPG debt)



Sources: Authority data and IMF staff calculations.

- The Lao P.D.R.'s domestic debt is estimated at LAK 23,234 billion or US$ 2,081 million at end-2021, which represents 13.9 percent of total PPG debt or 12.9 percent of GDP. The stock of domestic debt at end-2021 includes domestic bond issuance amounting to LAK 9.6 trillion (5.3 percent of GDP) to clear expenditure arrears in 2021 and LAK 5.4 trillion (3 percent of GDP) to recapitalize banks.[10] The stock of domestic debt does not include domestic debt held by SOEs (which is not explicitly guaranteed by the government) due to technical difficulties to consolidate the data by the authorities.

Text Table 3. Stock of Public Debt as of end-2021
(in percent of total external PPG and domestic debt)

At end-2021	as percent of total PPG debt	as percent of total external PPG debt	as percent of GDP	in mln USD
Total PPG debt	**100.0%**	**-**	**92.36%**	**14,950.8**
External debt	**86.1%**	**100.0%**	**79.5%**	**12,869.9**
ADB and IDA	10.9%	12.6%	10.02%	1,622.8
Other Multilateral	1.2%	1.3%	1.07%	172.7
China	36.9%	42.9%	34.07%	5,515.6
Other Bilateral	10.3%	12.0%	9.54%	1,544.6
Commercial Banks	6.4%	7.5%	5.96%	964.1
External Bonds	6.8%	7.9%	6.32%	1,023.1
Publicly Guaranteed	13.6%	15.8%	12.52%	2,027.1
Domestic debt	**13.9%**	**-**	**12.85%**	**2,080.8**

Sources: Authority data and IMF staff calculations

Note: $300 million from the PBoP swap and $130 million of deferred interest payment in 2021 is included in China, in addition to the authority numbers.

[10] The remaining government arrears as of end-2021, which can be calculated as a gap between the reported arrears and the domestic bond issuance to clear arrears, is LAK 926 billion (0.5 percent of GDP).

5. The government has addressed debt arrears by issuing domestic bonds, but there still exists a risk of domestic expenditure arrears going forward. The government reported LAK 10.5 trillion of public investment and recurrent expenditure arrears as of 2021, most of which was cleared by issuing domestic bonds corresponding to LAK 9,577 billion (5.3 percent of GDP) in 2021. Staff's estimates of the debt stock as of end-2022 include an additional LAK 23.5 trillion (11 percent of GDP) in arrears on old investment projects, which was not cleared by issuing domestic bonds as of end-2022. Given significant fiscal pressures and limited progress in terms of enhancing budget formulation, implementation, spending controls and monitoring, further domestic expenditure arrears are a risk going forward.

6. External debt repayment deferrals have taken place in 2020, 2021 and 2022: debt servicing to China was deferred during the past three years (US$ 220 million in 2020, US$ 450 million in 2021, and US$ 610 million in 2022), and US$150 million of deferred principal is supposed to be repaid in 2023. The deferred principal payments in 2020 and 2021 are included in 2021 debt stock in the debt bulletin. Staff includes the suspended interest payments (US$ 130 million in 2021 and US$ 190 million in 2022) in the debt stock for this DSA.

7. Despite eliminating the primary fiscal deficit, public debt increased in 2021, mainly because of exchange rate depreciation: Public sector debt is estimated at 92 percent of GDP by end 2021. The PPG debt stock increased by 16.4 percentage points of GDP from 2020 to 2021, largely driven by currency depreciation, the conversion of government arrears into domestic bonds, and domestic bond issuances to recapitalize banks. In addition, non-guaranteed SOE debts increased, in large part because of currency depreciation, adding to contingent liabilities.

8. The DSA estimates total private external debt to be 48.6 percent of GDP as of end-2021. Private external debt is estimated given the large numbers of PPP contracts.

RECENT MACROECONOMIC DEVELOPMENTS

9. This DSA is being conducted in the context of the 2023 Article IV Consultation. The previous Low-Income Country DSA (LIC-DSF) was considered by the Executive Board of the IMF in March 2022.

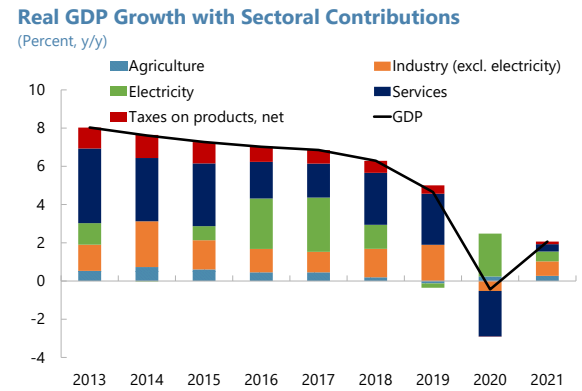
10. Macroeconomic vulnerabilities were aggravated by the impact of the COVID-19 pandemic.

- In the 2000s and 2010s, Lao P.D.R. had been among the fastest growing countries in the world. Growth of about 7 percent was driven by public and private investment, mainly for hydropower, mining, and transportation projects. The poverty rate fell substantially over the same period.[11] However, economic growth was partly fueled by public debt accumulation and did not support significant formal job creation. In fact, growth was already slowing prior to the COVID-19 shock,

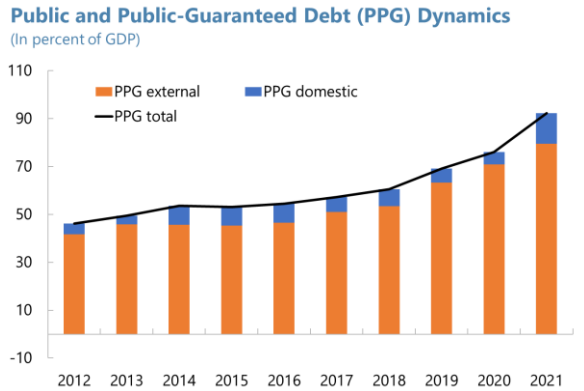
[11] World Bank estimates that headcount poverty fell to 18 percent by 2019. See 2022: *Lao People's Democratic Republic Poverty Assessment 2020: Catching Up and Falling Behind*. However, development needs remain high, as poverty rates are among the highest in the region and human capital levels are comparatively low.

as pressure from rising public debt had forced a slowdown in public investment. Other factors weighing on growth included the slowdown in credit growth and natural disasters.

- GDP growth is estimated to have been nearly flat in 2020 and just over 2 percent in 2021. COVID-19 containment measures included restrictions on movement and border closures. The economy had very limited market access since 2020 as concerns over public debt sustainability, slowing growth, low foreign exchange reserves, and tight foreign exchange liquidity grew.[12] The central bank provided loans to the government in 2020 (repaid in 2021) and at the end of 2021 and increased its financing to banks. The parallel market exchange rate depreciated significantly in mid-2021 and the exchange rate premium widened (going over 20 percent in 2021Q3), before narrowing to about 5 percent by the end of 2021 as the BoL allowed more official exchange rate flexibility from September 2021.



Sources: National authorities and IMF staff calculations and projections.



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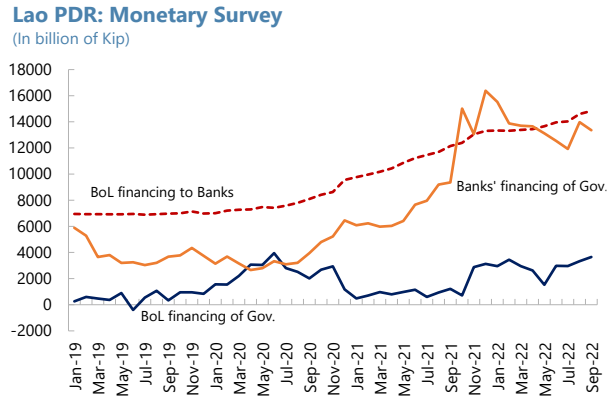


Sources: National authorities and IMF staff calculations.

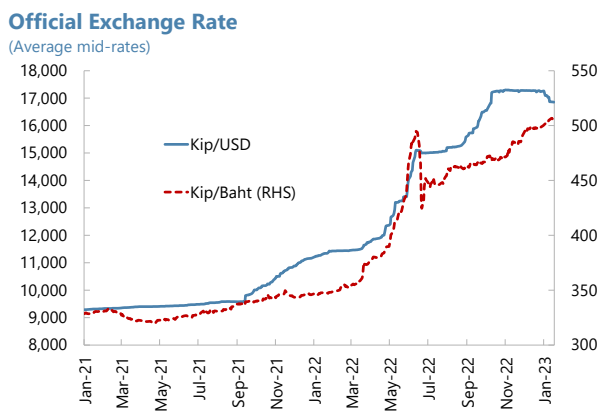
11. A gradual recovery was underway at the end of 2021, driven mainly by industry and agriculture. Services remained suppressed by COVID-19 containment measures. At the start of 2022, services picked

[12] Lao P.D.R.'s external market access has been largely confined to the Thai market.

up, as mobility restrictions were eased and the opening of the Lao-China high-speed railway and the Thanaleng Dry Port boosted transport and logistics sectors, in turn boosting wholesale and retail trade in the first half of 2022. Tourism also started to recover gradually. Exports overall had improved substantially during this period, growing by 26 percent (y/y) in 2021 compared to an average of 6 percent over the previous two years, contributing to a trade balance surplus.

12. The increase in global commodity prices in the first half of 2022 has accentuated pressures on the exchange rate from pre-existing macroeconomic balances, increasing inflation rates and substantially eroding living standards.

- The terms of trade pressures arising from the sharp increases in commodity prices in early 2022 accentuated concerns about FX liquidity and the public finances. The Lao kip depreciated by over 50 percent against the US dollar and over 40 percent against the Thai baht in 2022; the parallel exchange rate premia (against the USD) surged during the middle of 2022, before nearly closing by the end of the year.



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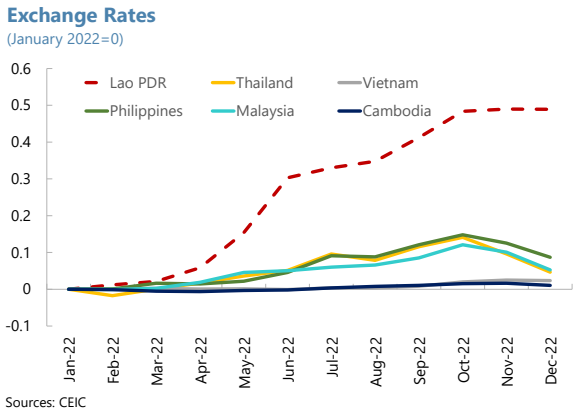
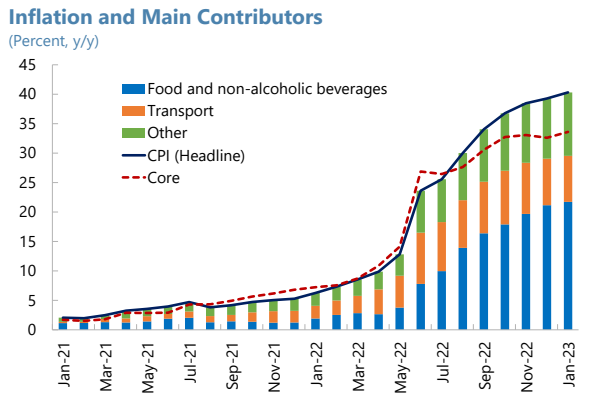


Sources: CEIC



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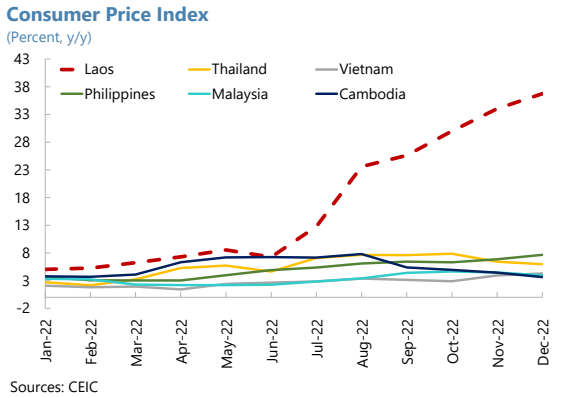

Sources: CEIC

- External shocks and exchange rate depreciation brought higher import prices, especially for fuel and fertilizers.[13] Inflation has surged, to 40 percent y/y as of January 2023; those items in the CPI basket with a larger share of imported goods have increased by considerably more.[14] Inflation in Lao P.D.R. is substantially higher than in neighboring countries; reflecting the depreciation but also non-tradeable price increases—as of November, all broad sectors have registered double-digit increases, which likely reflects a strong expectations channel over and above the effects of terms of trade and exchange rates; measures of core inflation have also increased.

- The inflation shocks severely stressed living standards. Fuel shortages were experienced in May 2022, reflecting a combination of the widening exchange rate premium and local fuel prices capped in kip based on the official exchange rate. The World Bank reports that 43 percent of households have experienced a decline in real incomes since the onset of the pandemic. Inflation has been highest on essential goods, resulting in cuts to food consumption, education, health care.[15] Poverty rates have increased.[16]

13. The economy has also faced weaker-than-expected global demand and tighter global financial conditions. Despite this, export growth has been strong, due to items such as paper goods products and perhaps reflecting the opening of the Lao-China railway and the Thanaleng Dry Port. But the interest rate differential against USD-denominated assets has worsened as global financial conditions have tightened and as inflation has surged by more in Lao P.D.R., likely exacerbating exchange rate pressures.

14. The authorities' eliminated the primary fiscal deficit in 2021. Fiscal consolidation has been substantial, driven equally by expenditure tightening (suspension of new public investment projects) and recovery in revenue collection, and by interest payment deferrals. (Note, however, the persistence and size of past expenditure arrears, which are not included in the fiscal accounts on a cash basis). The overall deficit improved by 4.3 percentage points of GDP in 2021, achieving a primary balance for the first time. Total revenue and grants reached 15 percent of GDP, closing the gap with pre-pandemic ratios.[17] Non-interest spending declined to 15.1 percent of GDP, down from 16.8 percent in 2020. Mid-year outturns for 2022 indicate that a primary surplus would have been attainable.

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[13] Oil prices increased by around 40 percent, while fertilizers increased by 70 percent in 2022 (y/y).

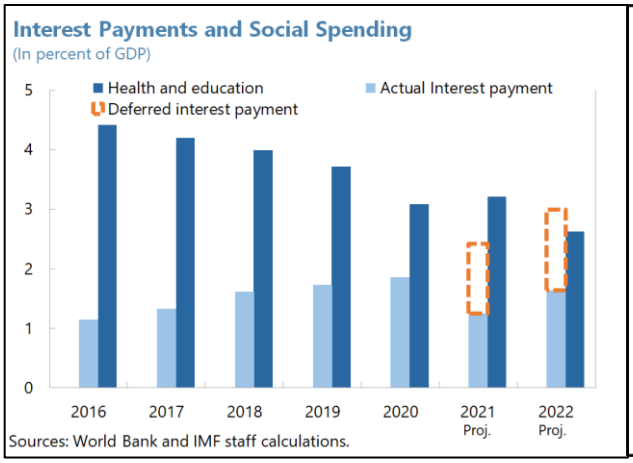
[14] Transport (16 percent of the CPI basket), increased by 50 percent YTD, food and non-alcoholic beverages (46 percent) by 47 percent YTD, and medical products by 48 percent.

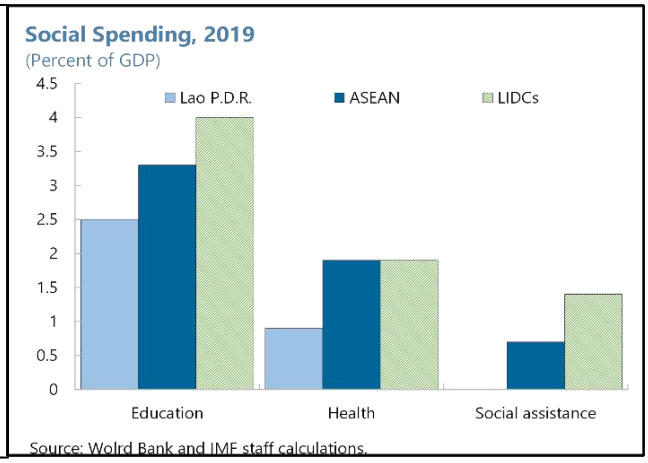
[15] World Bank (2022), *Monitoring Household Welfare in Lao PDR*, Report n.5, April-May

[16] Rising inflation could have pushed push an additional 2.9-3.7 percent of the Laotian population under the international poverty line (UNDP, 2022: *Addressing the Cost of Living Crisis in Developing Countries*). This is a significant increase from baseline: According to WB, the proportion of the Laotian population living in poverty was 18.3 percent in 2019 (World Bank, 2022: *Lao People's Democratic Republic Poverty Assessment 2020: Catching Up and Falling Behind*).

[17] At 25 percent YoY, nominal growth in fiscal revenues has far outstripped nominal GDP growth in 2021, exceeding the average historical elasticity (buoyancy). This performance can be explained by a combination of fiscal policy (active revenue mobilization efforts) and by the rebound from the historically low performance observed in 2020.

15. But spending cuts have come at the expense of important spending items and hence medium-term growth potential. The inability to protect those most affected by increases in the prices of essential goods has likely suppressed private consumption, while cuts to capital spending and health and education budgets hold back medium-term growth. Combined public spending on the social sectors (i.e., education, health, and social protection) is low compared with regional and income peers, and is well below international benchmarks (e.g., a recommendation of 4-6 percent of GDP for education). According to the MoF's Five-Year Plan, fiscal consolidation will rely heavily on spending austerity, and there is no plan to increase social benefits. As the cost of interest payments is increasing, significant primary surpluses are required to stabilize debt ratios in the medium term, which appears to be MoF's policy priority.





16. The Bank of the Lao P.D.R. (BoL) has acted to ease foreign exchange pressures. The BoL was lending to government in 2020 and was repaid in 2021. In 2021, it purchased government bonds from banks that were issued to cover expenditure arrears and to recapitalize banks.[18]

- The BoL has tightened domestic financial conditions: the BoL's extension of credit to banks and the government in 2022 has been fully offset by the issuance of BoL bonds.[19] The BoL has also increased its policy rate from 3.0 percent to 6.5 percent in 2022, and to 7.5 percent in February 2023, while also raising banks' reserve requirements.

[18] Domestic financing of the Ministry of Finance stood at LAK 16.7 trillion in 2021, including 1.9 trillion from the BoL. This figure includes 15 trillion of one-offs items (so-called "triangulation" and investment bonds, issued to cover arrears).

[19] In 2022, reserve money has grown strongly, mostly owing to the exchange rate depreciation changing the domestic currency value of foreign currency-denominated assets. By contrast, the BoL's domestic credit (NDA) growth has been much more subdued. From June 2021 to June 2022, the BoL's extension of credit to banks and the government has been completely offset by the issuance of BoL bonds to mop up liquidity. Banks' credit in Kip has picked up modestly, but remains subdued compared to historical average.

- It has tightened exchange controls,[20] while also allowing greater exchange rate flexibility.[21] A new FX law lays out surrendering requirements, but as yet secondary legislation to implement those provisions has not been implemented.

17. Despite nominal exchange rate depreciation, the external position in 2021 was still weak. The trade balance switched to a surplus in 2021-2022 with a sharp rise in exports, despite oil import pressures. Although the Covid-19 crisis had a significant impact on services exports (e.g., related to tourism), services imports also declined sharply, with the balance remaining somewhat stable. But despite the improvement in the current account position, the net international investment position—estimated to be -143 percent of GDP—has been deteriorating. Gross official reserves were down by US$ 170 million by end-August 2022 (from end-December 2021). Gross official reserves are projected to be US$ 1,101 million at end-2022, corresponding to 1.6 months of import coverage. For these reasons, the 2023 IMF Article IV Staff Report assesses the external position in 2021 to be substantially weaker than the level implied by fundamentals and desirable policies.

18. The financial sector remains weak. System-wide indicators conceal a wide range of capital levels and asset quality across banks. Capital levels at several of the largest banks are below requirements; these positions could yet be overstated because of loan forbearance.

MACROECONOMIC AND DEBT FORECASTS

19. The baseline outlook is for a period of stabilization, during growth rates are steady and inflation declines.

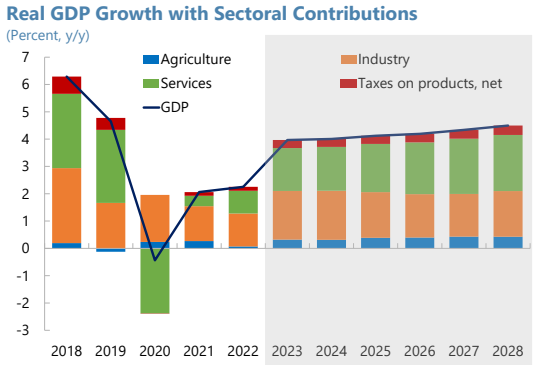
- The baseline assumes that there are no further external shocks, nor do risks manifest in the domestic economy. In particular, staff assumes that (i) for 2023, terms of trade stabilize after improving somewhat in the second half of 2022, allowing growth to improve, notwithstanding tight global financial conditions and external demand; (ii) improvement in services after the reopening of borders in 2022 and hence that tourism picks up substantially[22]; and (iii) fiscal policy and monetary policy continue to be tight, with no further lending from the BoL to government.

[20] Measures include daily limits on FX transactions, prioritizing access to FX, and closing FX bureaus.

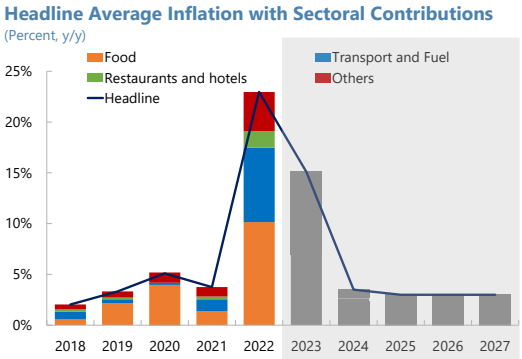
[21] The BoL has a mandate to maintain price stability, manage financial stability, and supervise both financial institutions and the payment system. The monetary policy framework is a mixed regime of monetary targeting and an exchange rate anchor. In September 2021, the exchange rate band was widened to ±1.5 percent. In October 2022, the band was further widened to ±4.5 percent.

[22] Pre-pandemic tourism was around 4 million visitors per year. At the time of writing, visitor numbers were estimated to have been 1.3 million in 2022, owing mainly to Thai and Vietnamese tourists and despite the lack of Chinese tourists.

- Under these assumptions, growth recovers, at 4.0 percent in 2023, gradually increasing to staff's estimate of long-run potential of 4.75 percent. Note that, with no fiscal space, public capital spending remains subdued, precluding a return to pre-pandemic growth rates.



Sources: National authorities and IMF staff calculations and projections.



Sources: National authorities and IMF staff calculations and projections.

- Average inflation is projected to fall, converging to a long-term rate of 3 percent by 2025. GDP deflator inflation is projected to be lower than CPI inflation in 2022, owing to negative terms of trade shocks in that year, and slightly higher than CPI inflation in 2023. Going forward, nominal GDP is expected to still grow at double digits in 2023 and stabilize in the 7-8 percent range thereafter.

Box 1. Potential Growth Assumptions

Although pre-crisis growth had been strong, growth accounting indicates that it was driven almost entirely by capital accumulation.[1] The strong investment was largely public, for major infrastructure and energy projects. The decomposition indicates that there was little benefit to the private economy, both in terms of productivity and employment growth. Indeed, net growth in formal private firms appears to have been negative, owing to lack of credit, labor mismatches, and competition from unregistered enterprises.

Going forward, employment is assumed to grow in line with UN working age population projections. TFP growth increases above historical rates due to efficiency gains, such as by improving the quality of public investment and facilitating private sector business environment in the long run [2]. Some increase in private investment is projected, but the state of the public finances, adverse financing conditions, and completion of major projects without firm plans for future projects reduce the scope for another surge in public investment. These assumptions imply a potential real growth rate of 4¾ percent.

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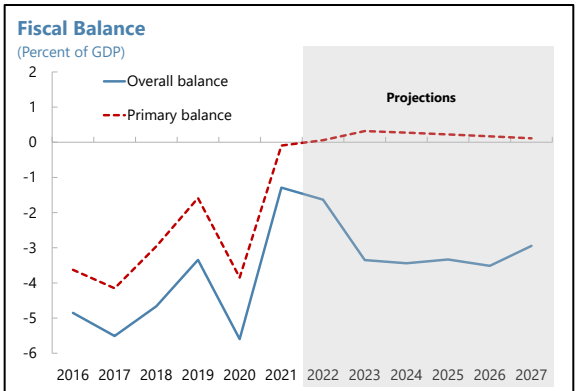
[1] A standard Cobb-Douglas production function with labor share of 60 percent was used. Data on output, capital stock, employment, and the labor share were obtained from the Penn World Tables 2021.
[2] The Government's National Agenda on Addressing Economic and Financial Difficulties aims to ease macroeconomic vulnerabilities and support a growth recovery. To support its implementation, the World Bank and the Asian Development Bank have recently prepared a Reform Roadmap that focuses on domestic resource mobilization, the quality of public spending, the governance of public and public-private investments, financial sector stability, and the business environment.

20. The improvement in fiscal balances is assumed to remain, with no further expenditure arrears.



Sources: National authorities and IMF staff calculations.

- Over the medium term, non-interest expenditure is expected to remain anchored at 2021 ratios to GDP (around 15 percent). A moratorium on new capital expenditure allows only existing projects to be completed. Ratios of wages, benefits and other spending to GDP are also expected to remain stable. Interest payments, however, are projected to increase significantly in the next few years, driving a rebound in overall expenditure (and the sizeable gap between primary and overall balances).

- The revenue gains seen in 2021 are expected to last, but the baseline assumes no further improvements. No additional measures on profit taxation are planned, hence their outturn is expected to be stable over the medium term; income tax and VAT outturns may suffer from compression in real incomes, and the reduction of the VAT rate from 10 to 7 percent in January 2022 will also undermine revenue mobilization. A portion of the increase in dividends observed in 2021 is assumed to be temporary and grants gradually decline, in line with historical trends. Overall, these losses are compensated by a small increase in other revenues (royalties, non-tax), leaving total inflows constant around 15 percent of GDP.

Text Table 4. Key Macroeconomic Assumptions

	2021	2022	2023	2024	2025	2026	2027	Long-term 1/
Real GDP (y/y growth)								
Current DSA	2.1	2.3	4.0	4.0	4.1	4.2	4.3	4.7
Previous DSA (2019 AVI)	6.7	6.8	6.8	6.8	6.7	6.5	6.6	5.9
GDP deflator (y/y growth)								
Current DSA	5.6	17.6	16.1	3.5	3.0	3.0	3.0	3.0
Previous DSA (2019 AVI)	3.1	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Primary fiscal balance (percent of GDP)								
Current DSA	-0.1	0.1	0.3	0.3	0.2	0.2	0.1	-0.3
Previous DSA (2019 AVI)	-2.4	-2.3	-2.2	-2.2	-2.2	-1.9	-1.8	-1.5
Revenue and grants (percent of GDP)								
Current DSA	15.0	14.9	15.1	15.1	15.1	15.1	15.0	14.7
Previous DSA (2019 AVI)	16.0	16.1	16.3	16.3	16.4	16.5	16.6	17.2
Primary expenditure (percent of GDP)								
Current DSA	15.1	14.8	14.8	14.8	14.9	14.9	14.9	15.0
Previous DSA (2019 AVI)	18.4	18.4	18.5	18.6	18.6	18.4	18.4	18.7
Fiscal balance (percent of GDP)								
Current DSA	-1.3	-1.6	-3.4	-3.5	-3.4	-3.6	-3.0	-2.6
Previous DSA (2019 AVI)	-3.9	-3.8	-3.7	-3.8	-3.7	-3.4	-3.3	-2.5
Current account balance (percent of GDP)								
Current DSA	-0.6	-6.0	-2.6	-6.2	-7.8	-8.4	-7.6	-6.1
Previous DSA (2019 AVI)	-11.1	-10.8	-10.9	-10.9	-10.8	-10.4	-9.7	-7.4
Exports of goods and services (percent of GDP)								
Current DSA	42.2	53.2	64.4	68.0	68.4	68.4	68.4	70.0
Previous DSA (2019 AVI)	36.5	37.0	37.1	35.8	35.3	34.8	34.4	31.9
Imports of goods and services (percent of GDP)								
Current DSA	38.2	52.8	60.0	66.1	69.0	70.3	70.9	75.2
Previous DSA (2019 AVI)	45.7	45.6	45.0	44.2	43.5	42.8	42.1	38.3

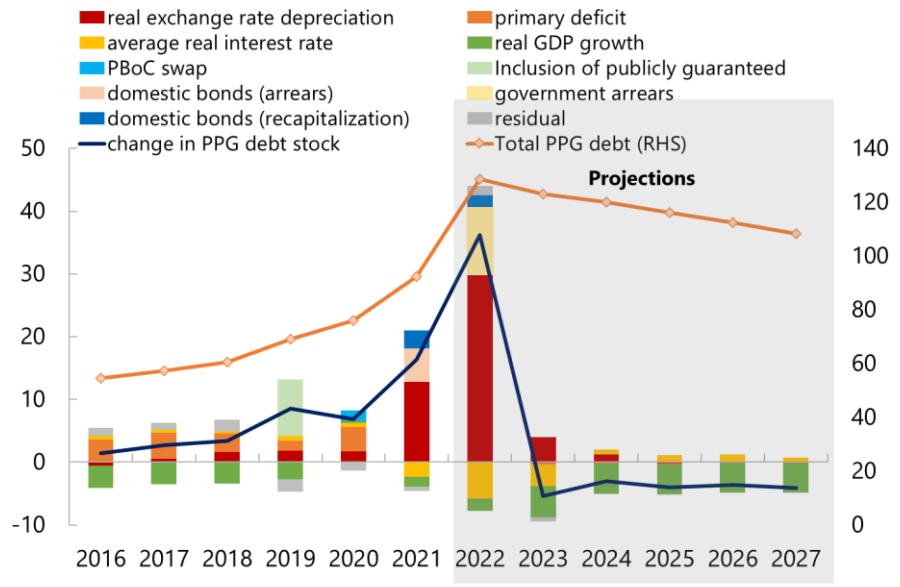
Sources: FSM authorities and IMF staff estimates and calculations.
1/ Average 2028-2042 for current DSA and 2028-2038 for previous one.

21. The PPG debt stock peaks in 2022 and declines thereafter, but remains high (Text figure 3 and Table 2).

- The debt ratio is estimated to have increased further, to 129 percent of GDP by end 2022. The main reason is exchange rate depreciation, contributing 30 percentage points to the estimated 36 percentage point increase over the year. A significant additional item is new government arrears to private contractors of around 11 percentage points of GDP, with bank recapitalization adding 2 percentage points, and deferred interest payments in 2022 adding 1 percentage point. In the absence of clear information except for the scheduled principal repayment of US$150 million in 2023, the remaining principal and interest servicing not made in 2020-2022 is assumed to be deferred until 2027, but is fully paid in the long run.[23]

- From 2022, the assumption that the government maintains a small primary surplus and positive GDP growth imply that debt gradually declines. Negative real interest rates contribute to the decline in 2023, as they did in 2022. However, debt is still above 100 percent until 2030 and around 60 percent in 2042.

Text Figure 3. PPG Debt Changes And Contributions
(In percent of GDP)

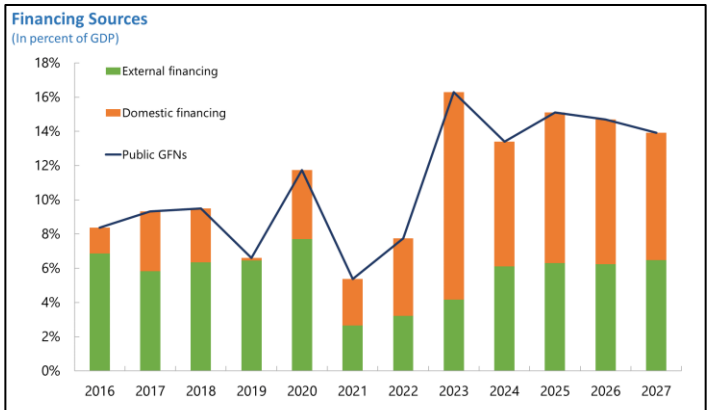


Sources: National authorities; and IMF staff calculations and projections.

22. Financing needs increase substantially from 2023; limited access to external financing implies reliance on domestic financing.

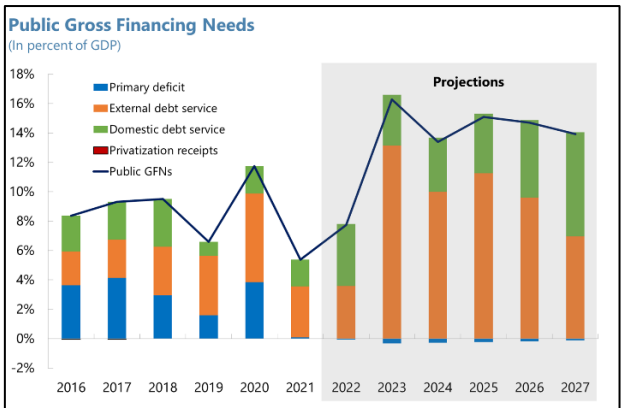
[23] The DSA assumes that for each external creditor, the total amortization is nearly equal to the initial debt stock, which comes from the authorities' debt bulletin. This means that the initial stock of debt from each external creditors will eventually be repaid in the future.

- Gross financing needs increase to nearly 16 percent of GDP in 2023, and remain at that level for several years. The increase arises mainly from external debt servicing that comes due in 2023 (including the swap line from the PBoC), which accounts for nearly 13 percentage points. Of this the majority is due to China. Smaller shares of external debt servicing are due to other bilateral creditors and to bondholders and commercial banks, with smaller share still to the ADB and World Bank.



Sources: National authorities; and IMF staff calculations and projections.

- External market access is assumed to only gradually be regained over the course of several years. Disbursements from commercial bank loans and bond issuance in international markets are assumed to be US$ 150 million in 2022 and 2023, about US$ 200 million from 2024 to 2026, and go up to US$ 500 million for 2027 and beyond. Hence the share of domestic financing would have to increase to nearly 12 percent of GDP in 2023. The share of domestic debt in total PPG public debt is expected to increase from 14 percent in 2021 to about 37 percent in 2030. Domestic banks are assumed in the baseline scenario as the main source of short- and medium- term domestic financing. Financing terms for external and domestic borrowings are described in Text Tables 5 and 6, respectively: the interest rate assumptions of domestic bonds are set in line with the recent historical issuance of investment/triangular bonds, which account for a large portion of MoF borrowing and debt stock (75 percent of the total debt stock) as of end-2021.[24]



Sources: National authorities; and IMF staff calculations and projections.

Decomposition of GFNs in 2023

	mln in Kip	mln in USD	percent of GDP
GFNs in 2023	42,733,278.8	2,295.1	16.3%
Primary deficit	-838,129.1	-45.0	-0.3%
Debt service for domestic existing debt	6,332,375.8	340.1	2.4%
Debt service for domestic new debt	2,717,302.0	145.9	1.0%
Debt service for external existing debt	34,185,509.2	1,836.1	13.0%
of which: ADB	1,222,412.4	65.7	0.5%
WB	807,256.0	43.4	0.3%
Other multilateral	294,998.3	15.8	0.1%
China (including PBoC swap)	20,642,494.0	1,108.7	7.9%
Principal	11,310,760.4	607.5	4.3%
Interest (excluding swap interest)	3,466,736.2	186.2	1.3%
Swap interest	279,285.6	15.0	0.1%
PBoC swap repayment	5,585,711.8	300.0	2.1%
Other bilateral	2,552,613.2	137.1	1.0%
Bonds/Commercial bank	8,665,735.3	465.4	3.3%
Debt service for external new debt	336,221.0	18.1	0.1%

Sources: Country authorities; and staff estimates and projections.

[24] Recent historical experience does not indicate changes in policy rates passing through to interest rates of government securities. Therefore, the DSA assumes no relationship between policy rates and bond interest rates.

Text Table 5. Loan Terms for Future External Disbursements

	Interest rate	Grace period	Maturity	Grant Element
Multilateral				
IDA (Blend)	2.0%	5	30	36%
AsDB	1.3%	8	32	45%
Others	2.0%	8	35	38%
Bilateral				
Paris Club	0.8%	9	30	51%
Non Paris Club (excluding China)	2.6%	6	23	24%
China	2.3%	6	20	26%
Commercial	6.9%	1	2	0%

Note. Disbursements from multilateral creditors other than the World Bank (IDA) and the Asian Development Bank ("Multilateral, Others") comprises less than 5 percent of disbursement amounts from multilateral creditors. Terms of these loans ("Others") have been kept identical to the previous 2019 DSA for lack of more information.

Text Table 6. Loan Terms for Future Domestic Loans

	interest rate				grace period	maturity	composition			
	2022-27	2028-32	2033-37	2038-42			2022-27	2028-32	2033-37	2038-42
Short-term debt										
T-bills (denominated in local currency)	4.3%	4.1%	4.1%	4.1%	0	1	2%	1%	1%	1%
MLT debt										
Denominated in local currency (LC)										
Bonds (1 to 3 years)-LC	4.6%	4.4%	4.4%	4.4%	1	3	8%	6%	3%	1%
Bonds (4 to 7 years)-LC	4.6%	4.4%	4.4%	4.4%	3	7	37%	27%	20%	15%
Bonds (4 to 7 years)-LC (with a higher interest rate)	4.9%	4.6%	4.6%	4.6%	3	7	53%	66%	76%	83%

Source: National authorities and IMF staff calculations.

Box 2. External Financing Assumptions

External financing assumptions tell a story of limited external access over the medium term, with a gradual pick-up as PPG debt ratios fall with higher growth.

Total external disbursement amounts average US$ 818 million from 2022 to 2027 and increase until 2042. External financing is made up of multilateral financing, bilateral financing, bonds, and commercial banks loans:

- Multilateral financing represents around 15 percent of the total external financing from 2022 to 2027, and is assumed to be around US$ 85 million on average from 2028 onward.
- Bond issuance is assumed to be US$ 150 million for 2022 and 2023 given the historical experience in Thai market. The bond issuance is assumed to continue at US$ 100 million every year from 2024.
- Commercial bank loans are assumed to be zero for 2022 and 2023, about US$ 100 million from 2024 to 2026, and US$ 400 million from 2027 onward.
- Other external financing is bilateral, representing 52 percent of the total external financing on average from 2022 to 2027 (on average: US$ 457 million), and 66 percent on average from 2028 to 2042, where it reaches US$ 1.7 billion.

All remaining financing needs are met through domestic borrowing.

23. The realism tools indicate that the baseline is consistent with Lao P.D.R.'s historical experiences and ensure the credibility of assumptions. The historical scenario does not divert largely from the baseline scenario, except for the present value of PPG external debt-to-GDP and -exports ratios, which increase explosively in the historical scenario after the mid-2020s. The explosive path of the historical scenario compared to the baseline comes from the large non-interest current account deficit in early 2010s, which is equivalent to 12.3 percent. Additionally, projected growth for 2022 in the baseline is not different significantly from the range of potential growth paths under various fiscal multipliers. Furthermore, the 3-year fiscal adjustment in the baseline is below the top quartile, which ensures the credibility of the baseline assumption.

24. The main risks arise from stresses to meet the government's financing needs. Assuming no further deferrals of debt servicing, that the authorities do not pursue monetary financing again, and in the absence of clear plans for other sources of financing (e.g. asset sales), domestic banks are assumed in the baseline scenario to lend to the government. Even without additional shocks, imposing loans to government implies significant crowding out of credit to the private sector: assuming broad money expands in line with nominal GDP growth, private sector credit growth would shrink to 3 percent y/y in 2023.

- There is a significant risk that the interest rate increases required to induce investors to finance government would be very large, increasing domestic financing costs. With weak bank balance sheets and negative real interest rates, domestic investors might demand shorter maturity bonds, increasing government rollover and refinancing risks. This might put pressure on the BoL to lend again to the government, without sterilization; in this case, reserve money would expand significantly, causing the exchange rate to depreciate by more than in the baseline, inflation to increase, and the kip value of liabilities (held by government and banks) to increase again. These pressures would be increased if the government were not able to sustain tax revenue gains and hence the primary surpluses assumed in the baseline.

- Deteriorating asset quality could trigger liquidity shortages and a solvency crisis in the banking system, causing a funding crisis for the government, both of which would cause the exchange rate and purchasing power to weaken further and domestic currency values of liabilities to increase.

- Even if the government were able to raise financing in local currency domestically, there remains the risk that the government would not be able to secure the foreign exchange to service external debt and to pay for imports without significant foreign exchange pressure.

25. The growth outlook is particularly uncertain, with risks tilted to the downside:

- It is possible that net trade improves further, given the potential for a faster pick-up in services (e.g., logistics and tourism) and remittances, which would bring valuable foreign exchange earnings.

- On the downside, in addition to the scenarios outlined in para 24, the external environment could be more adverse than expected in 2023. The low level of FX reserves and high debt burden imply that the economy has very little room to absorb external shocks; the result would likely be more pressure on the exchange rate, tighter FX liquidity, and higher inflation;

- Over the medium term, the economy risks long-term scarring: employment is yet to return to pre-pandemic levels, against the assumption in the baseline that employment rates pick up to fulfil the potential growth rate of 4.75 percent;[25] similarly, the assumption of a gradual increase in TFP might not be realized, given cuts to education and health spending. The country is highly vulnerable to natural disasters, particularly floods and droughts, which have significantly affected agriculture, forestry, water resources, food security, health, and economic growth.

COUNTRY CLASSIFICATION AND DETERMINATION OF STRESS TESTS

26. Debt-carrying capacity is assessed as weak. The country's Composite Indicator (CI) index of 2.15 indicates a weak debt-carrying capacity, based on October 2022 WEO and 2021 World Bank's Country Policy and Institutional Assessment (CPIA) (Text Table 7). Although the current CI is lower than the value in the previous 2019 DSA, which is 2.35, the indicative thresholds for external debt and benchmarks for public debt are unchanged: 30 percent for the present value (PV) of external debt-to-GDP ratio, 140 percent for the PV of external debt-to-exports ratio, 10 percent for the external debt service-to-exports ratio, 14 percent for the external debt service-to-revenue ratio, and 35 percent for the PV of public debt-to GDP ratio.

[25] World Bank (2022), *Lao PDR Economic Monitor: Restoring Macroeconomic Stability to Support Recovery*, April.

Text Table 7. Composite Indicator and Applicable Thresholds

Components	Coefficients (A)	Current DSA 10-year average values (B)	Current DSA CI Score components (A*B) = (C)	Current DSA Contribution of components	2019 DSA CI Score components (A*B) = (C)	2019 DSA Contribution of components
CPIA	0.385	3.038	1.17	54%	1.24	53%
Real growth rate (in percent)	2.719	3.657	0.10	5%	0.19	8%
Import coverage of reserves (in percent)	4.052	13.831	0.56	26%	0.51	21%
Import coverage of reserves^2 (in percent)	-3.990	1.913	-0.08	-4%	-0.06	-3%
Remittances (in percent)	2.022	0.453	0.01	0%	0.00	0%
World economic growth (in percent)	13.520	2.898	0.39	18%	0.48	20%
CI Score			2.15	100%	2.35	100%
CI rating			Weak		Weak	

Applicable Thresholds

APPLICABLE EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

APPLICABLE TOTAL public debt benchmark	
PV of total public debt in percent of GDP	35

Source: LIC DSF

27. Given the exposure to natural disasters and market financing, two tailored stress tests were added to the standard set of stress test scenarios.

- Lao P.D.R.'s exposure to natural disaster risks has increased in the past years. The 2018 disasters (two tropical cyclones and one dam collapse) have generated damages and losses estimated at 2 percent of GDP and recovery needs of around 3 percent of GDP. To mitigate future risk, the government of Lao P.D.R. has joined the regional catastrophe risk insurance pool. To account for Lao P.D.R.'s exposure to disasters, a natural disaster tailored stress test is added: a one-off shock of 3 percentage points to the external debt-to-GDP ratio in 2023 (the second year of the debt projection period), in line with the recent evidence and in accordance to estimates laid out in the 2019 DSA.[26]

- Lao P.D.R. has been accessing the Thai capital market since 2013 (although it temporarily lost market access in 2020) justifying the inclusion of a market financing shock occurring also in 2023. The calibrations include a 400-bps increase in the cost of new external commercial

[26] Lao P.D.R. is not included in the group of countries for which the climate disaster stress test scenario is automatically applied. As such, the default parameters, which are tailored to this group of countries, need to be adjusted when the stress test is applied to Lao P.D.R.

borrowing, a shortening of maturities by one-thirds and one-off foreign exchange depreciation by 15 percent.

28. Shocks related to contingent liabilities are incorporated in the contingent liabilities stress test using the tailored parameters. The contingent liability stress test applies a one-off 46.8 percentage point-increase in the debt-to-GDP ratio in the second year of the projection**.**

DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

29. The baseline scenario shows that three out of four sustainability indicators breach their thresholds from 2023 to 2027(Table 1 and Figure 1). Breaches are reported for both solvency (the present value (PV) of the debt-to-GDP ratio) and liquidity ratios (the debt service-to-export and the debt service-to-revenue ratios). The magnitude of breaches is significant (Text Table 8): for the PV of the debt-to-GDP and debt service-to-revenue ratios, the debt sustainability indicators continue above their thresholds during the next decade. The PV of the debt-to-GDP ratio reaches around 63 percent in 2022, and is estimated at about 40 percent in 2032. The external PPG debt service-to-revenue ratio peaks at 97 percent in 2023, and declines to 39 percent in 2032. The PPG debt service-to-exports ratio trends above or around its sustainability threshold (10 percent) from 2023 to 2027. Only the PV of the debt-to-exports ratio stays below its threshold (140 percent) over the entire projection period, with the highest value of 119 percent in 2022.

30. The stress tests indicate the vulnerability of Lao P.D.R.'s external debt dynamics to exports and currency depreciation and, to a lesser extent, to natural disasters and contingent liabilities (Table 3). An export shock generates the most severe deterioration of debt indicator for PV of debt-to-exports ratio and debt service-to-exports ratio, while the most extreme shock for PV of debt-to-GDP ratio and debt-service-to-revenue ratio is one-time depreciation.[27] Focusing on the extreme shocks, three of four sustainability indicators, except for the PV of debt-to-exports ratio, exceed the corresponding thresholds over the next decade.

[27] Export receipts grew by 26 percent (y/y) in 2021 contributing to a rare trade balance surplus. Exceptional export growth did not improve the FX position in 2021 as only one third of export proceeds entered the domestic banking sector (according to information provided by the authorities).

Text Table 8. Magnitude of Breaches
(In percent of thresholds/benchmark)

	Average breach per period (in percent of threshold/Benchmark)		
	2022-2024	**2025-2027**	**2028-2032**
External PPG Debt			
PV of debt-to-GDP ratio	106.3%	65.4%	39.4%
PV of debt-to-exports ratio		no breach	
Debt service-to-exports ratio	75.6% 1/	35.6%	5.7% 2/
Debt service-to-revenue ratio	368.5%	378.0%	226.6%
Total PPG Debt			
PV of debt-to-GDP ratio	176.7%	151.1%	121.5%

1/ For 2022-2024, the calculation is the average of 2023 and 2024 because of no breach in 2022.
2/ For 2028-2032, the table displays the breach in 2029 because of no breaches for other years.
Sources: IMF staff calculations

B. Public Sector Debt Sustainability Analysis

31. The sustainability indicator for total PPG debt also breaches its threshold significantly over the next decades (Table 2 and Figure 2). The PV of the debt-to-GDP ratio exceeds its benchmark (35 percent) by a significant amount over the entire projection period. The debt service-to-revenue and grants ratio shows a rapid hike from 52 percent in 2022 to 110 percent in 2023 and goes down to 83 percent in 2032**.**

32. Public debt is most vulnerable to the contingent liabilities shock, which is driven by SOE and PPP risks and a financial market shock (Figure 2 and Table 4**)**. The PV of public debt-to-GDP ratio under the shock reaches 142 percent in 2023 and the debt service-to-revenue and grants would peak at 168 percent in 2029.

RISK RATING AND VULNERABILITIES

33. The Lao P.D.R. is assessed to be in external and overall debt distress. The assessment reflects the on-going negotiations about debt service deferral and the significant debt vulnerabilities emanating from the large and protracted threshold breaches of most solvency and liquidity indicators.

34. Overall debt is assessed to be unsustainable under the current baseline. Due to the high debt service ratios and critical macroeconomic conditions (including low foreign exchange reserves and liquidity, limited external market access since 2020, high dependence on domestic financing to close fiscal financing gap, and no fiscal space), overall debt is assessed to be unsustainable under the current baseline.

35. Downside risks would add further stress.

- The gross financing needs increase sharply in 2023 and exceed the 14 percent benchmark value.[28] On the assumption that external financing remains limited, the scale of domestic financing is substantial and presents a major stress to the projections. In the absence of confirmed alternative sources of funding, the baseline assumes that domestic banks lend to government. At the least, this would imply significant crowding out of credit availability to the private sector, stressing the recovery. The main risk is that the interest rate increases needed to induce the private sector to finance the government would be significantly larger than in the baseline, reducing growth, increasing government financing costs, and increasing rollover risks. Recourse to the BoL to finance the deficit would put further pressure on the exchange rate, increasing the domestic currency value of external liabilities to government and banks again.

- The economy continues to be vulnerable to external shocks. Globally, inflation may prove to be more persistent than anticipated, leading to tighter financial conditions.

- A manifestation of contingent liability risks would greatly stress the public finances. Vulnerabilities shown by the DSA include: shocks to exports, depreciation of the currency, large exposure to PPPs, and recapitalization needs of the banking sector, and the impact of natural disasters. The low level of reserves, a shallow domestic debt market, and limited access to capital markets add to these vulnerabilities.

In all these cases, liquidity pressures would increase, whether because financial inflows slow or because of domestic financing pressures, leading to solvency pressures in the banking system and causing funding stresses for the government and/or begetting monetization or financial repression that would cause the parallel exchange rate gap to widen again;[29]

36. Debt would remain at very high levels even with higher growth and revenue collection. The underlying problem is the starting point of debt:

- Alternative scenarios indicate that debt would remain high for the next 10 years, even assuming growth reaches the authorities' targets, which are much higher than the baseline, and with primary surpluses persisting for the duration of the simulation.

- The primary surpluses required to reduce debt below risk thresholds in the DSA within a few years (without harming long-term development prospects and without debt restructuring) are not socially feasible. Staff estimates that to reduce PPG debt to around 60 percent of GDP—a level that would still be somewhat risky for a heavily dollarized low-income economy—by 2027, the government would have to run a primary surplus of about 10 percent of GDP on average

[28] There is no Emerging Market Bond Index (EMBI) or equivalent spread for Lao P.D.R., which impedes an assessment of the second benchmark related to market sentiment.

[29] For example, if banks were to take significantly more "triangulation" bonds (in effect, IOUs from the government) onto their balance sheets, their solvency would be stressed. Liquidity would improve by allowing those bonds to be used as collateral at the BoL, but at the cost of expanding the money base.

each year for five years. Nonetheless, a commitment to a sustained revenue-led fiscal consolidation would improve market perceptions, reducing pressures to finance domestically.

37. The government recognizes the need to bring down public debt in its 9th Five-Year Plan and the National Agenda for addressing economic and financial difficulties. The government has responded to these challenges with measures to improve the public finances and to ease foreign exchange pressures. The government has announced this year that it has sought external debt reprofiling. The government has reached a small primary surplus, following substantial fiscal consolidation driven equally by expenditure tightening (both recurrent and suspension of new public investment projects) and recovery in revenue collection, and by interest payment deferrals. The Bank of the Lao P.D.R. (BoL) has allowed the exchange rate to depreciate, in line with staff advice, and has also attempted to reduce exchange pressure by soaking up excess kip liquidity and tightening rules on exchange markets.

38. A comprehensive set of adjustment policies is needed that recognizes the full extent of the macroeconomic imbalances, based on realistic expectations of revenue capacity and the time for reforms to show their full benefits.

- *Fiscal policy:* A preferred growth-friendly fiscal consolidation would be based mainly on revenue mobilization, eliminating tax exemptions and VAT rate reductions, improving tax administration, and broadening the tax base. Spending should be directed to social and development needs, with heightened attention to financial management.

- *Monetary policy:* The BoL should continue with bond issuances and interest rate increases, to ease pressures on the exchange rate and reduce pressures on bank solvency and to facilitate foreign exchange inflows.

- *Structural reforms* are needed to complement efforts to restore the public finances to health and improve external balances and put the economy on a structural transformation path for sustained and inclusive growth. With fiscal restraint required for a sustained period, growth will have to come from the private sector. Historically, firms have been held back by bureaucracy, difficult access to credit, labor market mismatches, and competition from unregistered enterprises. Firms need improved governance—in particular, simpler regulation, consistently implemented, and anti-corruption measures.

AUTHORITIES' VIEWS

39. The authorities affirmed their intention to service debt repayments in full in 2023. Financing would be mostly domestic, with more tax revenue raised, sales of mining and hydroelectric concessions and fees, and borrowing from commercial banks and domestic issuance of bonds. A small share would be issued externally. The Lao government has discussed with China holding debt servicing during 2020, 2021 and 2022 until negotiations about postponements has been completed. There is not yet a schedule on when deferred amounts would be repaid. The government is in negotiations with bilateral creditors about future repayments, but no details were available at the time of the Article IV consultation. The authorities

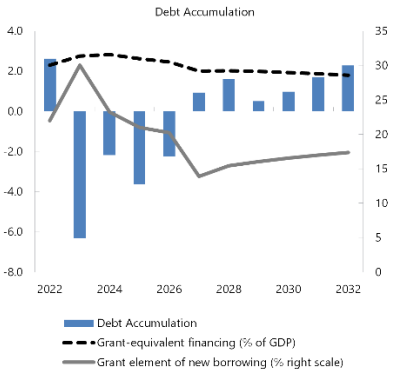
noted that, under Lao law, government arrears to the private sector were not counted until a bond was issued.

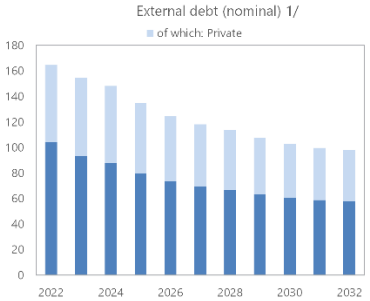
Table 1. External Debt Sustainability Framework, Baseline Scenario, 2021–2042

(In percent of GDP, unless otherwise)

	Actual	Projections								Average 8/	
	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
External debt (nominal) 1/	**128.1**	**164.8**	**154.5**	**148.2**	**135.0**	**124.5**	**118.3**	**97.8**	**85.2**	**101.1**	**124.3**
of which: public and publicly guaranteed (PPG)	**79.5**	**104.5**	**93.5**	**87.6**	**79.6**	**73.4**	**69.7**	**57.7**	**50.2**	**54.3**	**74.2**
Change in external debt	8.7	36.6	-10.3	-6.3	-13.2	-10.5	-6.2	-2.0	-1.9		
Identified net debt-creating flows	**-5.3**	**-1.2**	**-9.9**	**-6.9**	**-4.7**	**-3.6**	**-4.1**	**-5.5**	**-2.0**	**1.2**	**-5.8**
Non-interest current account deficit	**-1.2**	**3.6**	**-3.8**	**0.2**	**2.7**	**4.1**	**4.4**	**3.4**	**5.0**	**12.3**	**2.0**
Deficit in balance of goods and services	-4.0	-0.4	-4.4	-1.9	0.6	1.9	2.5	4.5	3.1	**13.1**	**1.0**
Exports	42.2	53.2	64.4	68.0	68.4	68.4	68.4	69.2	55.7		
Imports	38.2	52.8	60.0	66.1	69.0	70.3	70.9	73.8	58.7		
Net current transfers (negative = inflow)	-1.2	-1.6	-2.2	-2.2	-2.2	-2.0	-2.0	-1.2	-0.7	**-1.8**	**-1.8**
of which: official	-1.4	-1.3	-1.4	-1.4	-1.4	-1.3	-1.3	-0.6	-0.3		
Other current account flows (negative = net inflow)	4.0	5.7	2.7	4.3	4.2	4.2	3.9	0.0	2.6	**1.0**	**2.8**
Net FDI (negative = inflow)	**-5.8**	**-3.7**	**-5.4**	**-6.8**	**-6.8**	**-6.8**	**-6.6**	**-6.0**	**-4.3**	**-6.9**	**-6.1**
Endogenous debt dynamics 2/	**1.6**	**-1.2**	**-0.6**	**-0.3**	**-0.7**	**-1.0**	**-1.9**	**-2.8**	**-2.7**		
Contribution from nominal interest rate	1.8	2.3	6.5	6.0	5.1	4.4	3.2	1.6	1.2		
Contribution from real GDP growth	-2.5	-3.5	-7.1	-6.3	-5.8	-5.3	-5.1	-4.4	-3.9		
Contribution from price and exchange rate changes	2.3	...	...	...	...	...	...	...	...		
Residual 3/	**14.1**	**37.8**	**-0.4**	**0.6**	**-8.5**	**-6.9**	**-2.1**	**3.5**	**0.1**	**3.6**	**3.1**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**49.7**	**63.3**	**61.9**	**60.4**	**53.8**	**48.5**	**46.5**	**39.7**	**38.0**		
PV of PPG external debt-to-exports ratio	**117.7**	**119.0**	**96.1**	**88.8**	**78.7**	**70.9**	**68.0**	**57.4**	**68.2**		
PPG debt service-to-exports ratio	**8.3**	**6.7**	**20.4**	**14.7**	**16.5**	**14.0**	**10.2**	**7.9**	**7.8**		
PPG debt service-to-revenue ratio	**26.8**	**26.9**	**96.9**	**73.0**	**81.4**	**69.3**	**50.1**	**38.8**	**30.2**		
Gross external financing need (Million of U.S. dollars)	513.8	1721.6	2629.6	2564.8	3140.4	3121.1	2685.1	2068.9	2217.8		
Key macroeconomic assumptions											
Real GDP growth (in percent)	2.1	2.3	4.0	4.0	4.1	4.2	4.3	4.8	4.8	**5.7**	**4.2**
GDP deflator in US dollar terms (change in percent)	-1.9	-19.2	-11.4	-4.9	1.3	2.0	1.9	1.9	1.9	**1.8**	**-1.9**
Effective interest rate (percent) 4/	1.5	1.5	3.6	3.8	3.7	3.4	2.7	1.7	1.4	**2.3**	**2.7**
Growth of exports of G&S (US dollar terms, in percent)	21.0	4.2	11.5	4.4	6.0	6.4	6.2	7.3	0.0	**8.3**	**6.7**
Growth of imports of G&S (US dollar terms, in percent)	8.3	14.2	4.6	9.0	10.2	8.4	7.1	8.7	0.0	**4.8**	**8.3**
Grant element of new public sector borrowing (in percent)	...	22.0	30.0	23.2	21.0	20.2	13.9	17.4	19.2	**...**	**19.3**
Government revenues (excluding grants, in percent of GDP)	13.0	13.3	13.6	13.7	13.8	13.9	13.9	14.1	14.3	**14.6**	**13.8**
Aid flows (in Million of US dollars) 5/	377.1	433.6	578.3	403.0	296.4	232.7	219.3	240.2	310.5		
Grant-equivalent financing (in percent of GDP) 6/	...	2.3	2.8	2.8	2.6	2.5	2.0	1.8	1.5	**...**	**2.2**
Grant-equivalent financing (in percent of external financing) 6/	...	47.9	48.5	37.5	34.5	33.1	26.4	24.0	26.2	**...**	**32.2**
Nominal GDP (Million of US dollars)	18,533	15,304	14,091	13,940	14,701	15,631	16,623	22,918	44,123		
Nominal dollar GDP growth	0.1	-17.4	-7.9	-1.1	5.5	6.3	6.3	6.8	6.8	**7.7**	**2.3**
Memorandum items:											
PV of external debt 7/	98.3	123.6	122.9	121.0	109.3	99.6	95.1	79.9	72.9		
In percent of exports	232.9	232.2	190.7	177.9	159.8	145.6	139.0	115.4	131.1		
Total external debt service-to-exports ratio	23.1	21.2	43.3	36.8	37.2	33.1	26.9	16.8	7.8		
PV of PPG external debt (in Million of US dollars)	9205.6	9690.3	8725.6	8417.7	7912.7	7583.1	7729.6	9110.1	16759.4		
(PVt-PVt-1)/GDPt-1 (in percent)		2.6	-6.3	-2.2	-3.6	-2.2	0.9	2.3	1.9		
Non-interest current account deficit that stabilizes debt ratio	-9.9	-33.0	6.4	6.5	15.9	14.6	10.6	5.4	7.0		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

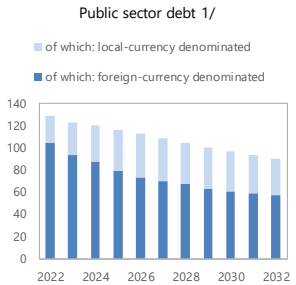
8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

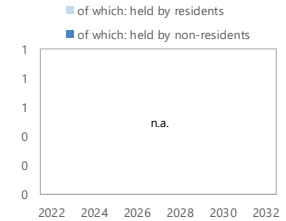
Table 2. Public Sector Debt Sustainability Framework, Baseline Scenario, 2021–2042

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
Public sector debt 1/	**92.4**	**128.5**	**123.1**	**120.1**	**116.0**	**112.4**	**108.3**	**89.9**	**63.6**	**61.2**	**108.5**
of which: external debt	**79.5**	**104.5**	**93.5**	**87.6**	**79.6**	**73.4**	**69.7**	**57.7**	**50.2**	**54.3**	**74.2**
Change in public sector debt	16.4	36.2	-5.4	-3.0	-4.0	-3.6	-4.1	-3.4	-2.0		
Identified debt-creating flows	**8.9**	**-7.8**	**-8.8**	**-4.2**	**-3.9**	**-3.6**	**-4.1**	**-3.5**	**-2.3**	**1.5**	**-4.6**
Primary deficit	**0.1**	**-0.1**	**-0.3**	**-0.3**	**-0.2**	**-0.2**	**-0.1**	**0.2**	**0.5**	**2.8**	**-0.1**
Revenue and grants	15.0	14.9	15.1	15.1	15.1	15.1	15.0	14.7	14.8	**17.7**	**14.9**
of which: grants	2.0	1.6	1.5	1.4	1.3	1.2	1.1	0.6	0.5		
Primary (noninterest) expenditure	15.1	14.8	14.8	14.8	14.9	14.9	14.9	14.9	15.3	**20.4**	**14.9**
Automatic debt dynamics	**8.8**	**-7.7**	**-8.4**	**-4.0**	**-3.6**	**-3.4**	**-4.0**	**-3.7**	**-2.8**		
Contribution from interest rate/growth differential	-3.9	-7.7	-8.4	-4.0	-3.6	-3.4	-4.0	-3.7	-2.8		
of which: contribution from average real interest rate	-2.4	-5.7	-3.5	0.8	1.1	1.3	0.7	0.5	0.2		
of which: contribution from real GDP growth	-1.5	-2.0	-4.9	-4.7	-4.8	-4.7	-4.7	-4.2	-3.0		
Contribution from real exchange rate depreciation	12.7	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**7.5**	**44.0**	**3.3**	**1.2**	**-0.2**	**0.0**	**0.0**	**0.1**	**0.3**	**3.4**	**4.4**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**69.7**	**100.9**	**96.1**	**93.6**	**90.5**	**87.8**	**85.4**	**72.2**	**51.5**		
PV of public debt-to-revenue and grants ratio	**463.7**	**677.1**	**636.9**	**619.4**	**598.5**	**582.6**	**568.7**	**491.7**	**348.6**		
Debt service-to-revenue and grants ratio 3/	**35.1**	**52.4**	**110.1**	**90.5**	**101.3**	**98.7**	**93.5**	**83.4**	**49.9**		
Gross financing need 4/	5.4	7.7	16.3	13.4	15.1	14.7	13.9	12.5	7.9		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	2.1	2.3	4.0	4.0	4.1	4.2	4.3	4.8	4.8	**5.7**	**4.2**
Average nominal interest rate on external debt (in percent)	1.0	1.3	2.7	2.8	2.6	2.5	1.7	2.1	2.4	**2.2**	**2.1**
Average real interest rate on domestic debt (in percent)	-0.8	-12.4	-9.6	1.4	1.9	2.4	2.2	1.5	1.5	**3.2**	**-0.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	19.0	...	...	...	...	...	...	...	...	**1.7**	**...**
Inflation rate (GDP deflator, in percent)	5.6	17.6	16.1	3.5	3.0	3.0	3.0	3.0	3.0	**3.8**	**5.6**
Growth of real primary spending (deflated by GDP deflator, in percent)	-8.3	0.3	3.4	4.5	4.6	4.2	4.3	4.8	5.0	**3.8**	**4.0**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-16.3	-36.2	5.1	2.7	3.8	3.4	4.0	3.7	2.5	**-8.7**	**0.2**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**Yes**





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, government-guaranteed debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

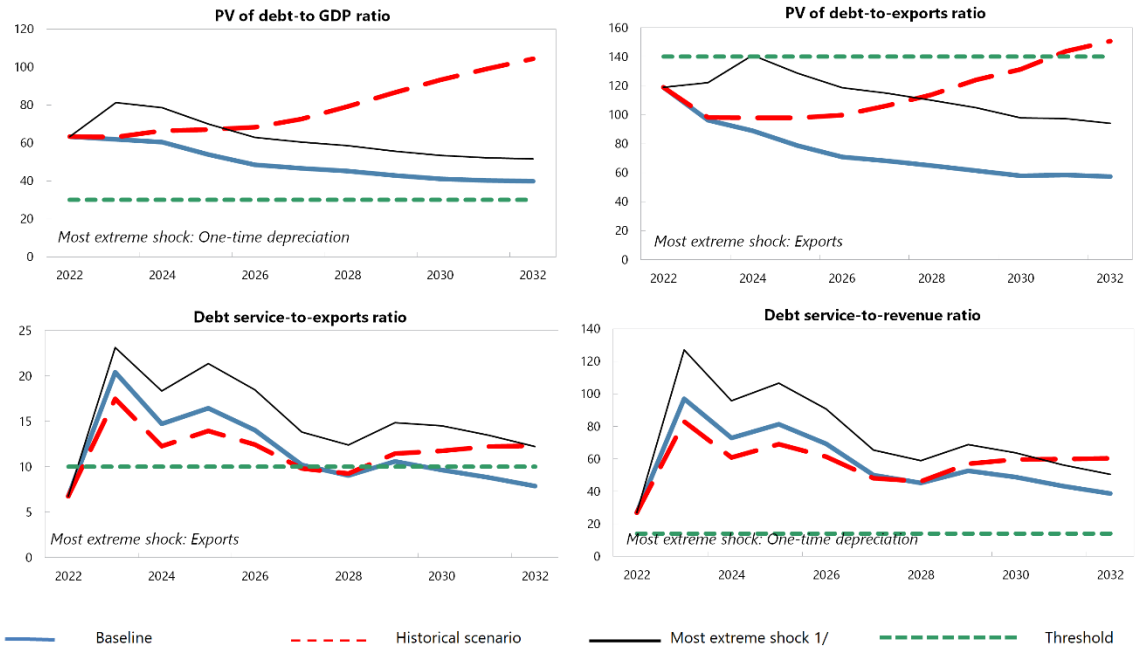
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Indicators Of Public And Publicly Guaranteed External Debt Under Alternative Scenarios, 2022–2032



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	Yes	No
Commodity price	n.a.	n.a.
Market financing	No	No

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

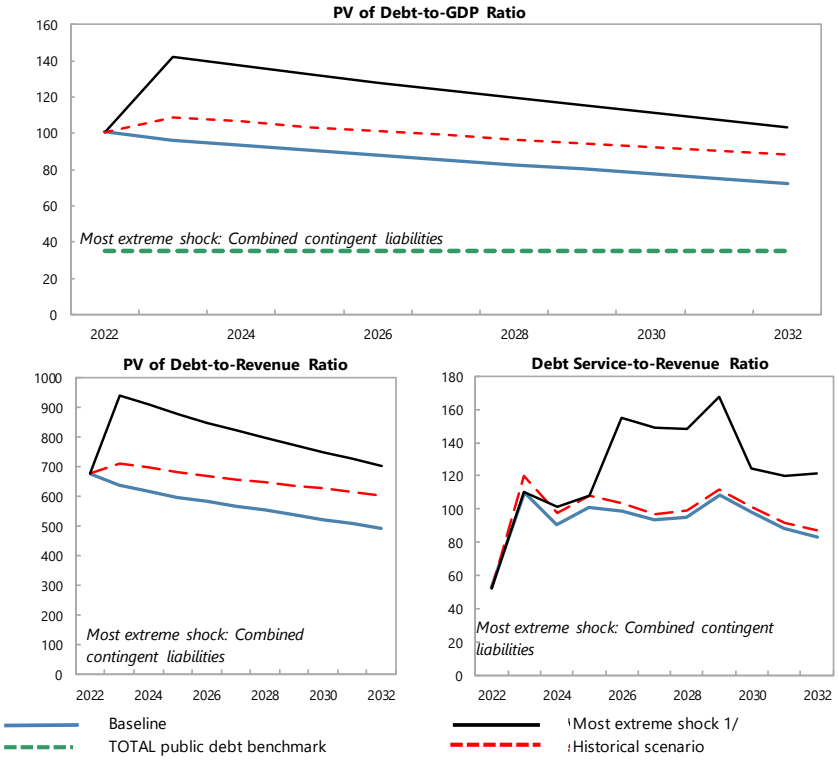
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	3.8%	5.0%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	18	20
Avg. grace period	4	5

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Indicators Of Public Debt Under Alternative Scenarios, 2022–2032



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	42%	20%
Domestic medium and long-term	57%	79%
Domestic short-term	1%	1%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	3.8%	3.8%
Avg. maturity (incl. grace period)	18	18
Avg. grace period	4	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	-0.9%	-0.9%
Avg. maturity (incl. grace period)	6	6
Avg. grace period	2	2
Domestic short-term debt		
Avg. real interest rate	-1.4%	-1.4%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2022–2032

(In percent)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio											
Baseline	63	62	60	54	49	46	45	43	41	40	40
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	**63**	**63**	**66**	**67**	**68**	**73**	**79**	**86**	**93**	**99**	**104**
B. Bound Tests											
B1. Real GDP growth	**63**	**65**	**66**	**59**	**53**	**51**	**50**	**47**	**45**	**44**	**44**
B2. Primary balance	**63**	**63**	**63**	**56**	**51**	**49**	**48**	**45**	**44**	**43**	**42**
B3. Exports	**63**	**70**	**81**	**74**	**68**	**66**	**65**	**62**	**59**	**56**	**55**
B4. Other flows 3/	**63**	**64**	**65**	**58**	**53**	**51**	**49**	**47**	**45**	**44**	**43**
B5. Depreciation	**63**	**81**	**79**	**70**	**63**	**60**	**59**	**56**	**53**	**52**	**52**
B6. Combination of B1-B5	**63**	**67**	**69**	**63**	**57**	**55**	**54**	**51**	**49**	**47**	**46**
C. Tailored Tests											
C1. Combined contingent liabilities	**63**	**70**	**69**	**63**	**58**	**57**	**57**	**55**	**54**	**53**	**53**
C2. Natural disaster	**63**	**63**	**62**	**55**	**50**	**48**	**47**	**45**	**43**	**42**	**42**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**63**	**70**	**69**	**60**	**54**	**51**	**50**	**47**	**45**	**44**	**44**
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	119	96	89	79	71	68	65	61	58	58	57
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	119	98	98	98	100	106	114	124	131	**144**	**151**
B. Bound Tests											
B1. Real GDP growth	119	96	89	79	71	68	65	61	58	58	57
B2. Primary balance	119	97	92	82	74	71	68	65	61	62	61
B3. Exports	119	122	**141**	128	119	115	110	105	98	97	94
B4. Other flows 3/	119	99	95	85	77	74	71	67	63	63	62
B5. Depreciation	119	96	88	78	70	67	64	61	57	58	57
B6. Combination of B1-B5	119	114	100	98	90	86	83	78	73	73	72
C. Tailored Tests											
C1. Combined contingent liabilities	119	109	102	91	85	84	82	80	76	77	76
C2. Natural disaster	119	100	93	82	74	72	68	65	61	62	61
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	119	96	89	77	69	66	63	60	56	57	56
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	7	**20**	**15**	**16**	**14**	**10**	9	**11**	10	9	8
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	7	**17**	**12**	**14**	**12**	10	9	**11**	**12**	**12**	**12**
B. Bound Tests											
B1. Real GDP growth	7	**20**	**15**	**16**	**14**	**10**	9	**11**	10	9	8
B2. Primary balance	7	**20**	**15**	**17**	**14**	**10**	9	**11**	10	9	8
B3. Exports	7	**23**	**18**	**21**	**18**	**14**	**12**	**15**	**15**	**13**	**12**
B4. Other flows 3/	7	**20**	**15**	**17**	**14**	**10**	9	**11**	**10**	9	8
B5. Depreciation	7	**20**	**15**	**16**	**14**	**10**	9	**11**	10	9	8
B6. Combination of B1-B5	7	**22**	**17**	**19**	**16**	**12**	**10**	**13**	**12**	**11**	10
C. Tailored Tests											
C1. Combined contingent liabilities	7	**20**	**15**	**17**	**15**	**11**	10	**11**	**10**	10	9
C2. Natural disaster	7	**21**	**15**	**17**	**15**	**11**	9	**11**	**10**	9	8
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	7	**20**	**16**	**19**	**15**	8	9	**10**	10	9	8
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	**27**	**97**	**73**	**81**	**69**	**50**	**45**	**53**	**49**	**43**	**39**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	**27**	**83**	**61**	**69**	**61**	**48**	**46**	**57**	**60**	**60**	**60**
B. Bound Tests											
B1. Real GDP growth	**27**	**102**	**80**	**89**	**76**	**55**	**50**	**58**	**54**	**47**	**42**
B2. Primary balance	**27**	**97**	**73**	**82**	**70**	**51**	**46**	**54**	**50**	**45**	**40**
B3. Exports	**27**	**97**	**77**	**89**	**77**	**57**	**52**	**62**	**62**	**56**	**51**
B4. Other flows 3/	**27**	**97**	**74**	**83**	**71**	**52**	**47**	**55**	**52**	**46**	**41**
B5. Depreciation	**27**	**127**	**96**	**106**	**91**	**65**	**59**	**69**	**64**	**56**	**50**
B6. Combination of B1-B5	**27**	**96**	**77**	**86**	**73**	**54**	**49**	**58**	**55**	**49**	**44**
C. Tailored Tests											
C1. Combined contingent liabilities	**27**	**97**	**76**	**84**	**72**	**53**	**48**	**56**	**52**	**47**	**42**
C2. Natural disaster	**27**	**97**	**73**	**82**	**69**	**50**	**45**	**53**	**49**	**44**	**39**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**27**	**97**	**81**	**92**	**72**	**40**	**45**	**52**	**48**	**43**	**38**
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Sensitivity Analysis for Key Indicators of Public Debt, 2022–2032

(In percent)

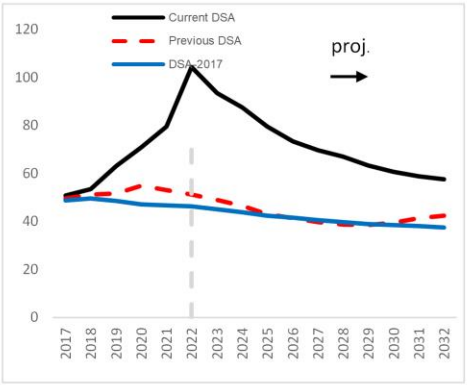
	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
	PV of Debt-to-GDP Ratio										
Baseline	**101**	**96**	**94**	**91**	**88**	**85**	**83**	**80**	**78**	**75**	**72**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	**101**	**108**	**107**	**104**	**101**	**99**	**97**	**95**	**93**	**91**	**88**
B. Bound Tests											
B1. Real GDP growth	**101**	**101**	**104**	**102**	**100**	**98**	**96**	**94**	**92**	**90**	**88**
B2. Primary balance	**101**	**100**	**103**	**99**	**96**	**94**	**91**	**88**	**85**	**82**	**79**
B3. Exports	**101**	**104**	**113**	**110**	**107**	**104**	**101**	**98**	**94**	**91**	**87**
B4. Other flows 3/	**101**	**98**	**98**	**95**	**92**	**90**	**87**	**84**	**81**	**78**	**76**
B5. Depreciation	**101**	**105**	**101**	**97**	**93**	**89**	**86**	**82**	**78**	**75**	**71**
B6. Combination of B1-B5	**101**	**97**	**100**	**97**	**94**	**92**	**89**	**86**	**83**	**81**	**78**
C. Tailored Tests											
C1. Combined contingent liabilities	**101**	**142**	**138**	**133**	**128**	**124**	**120**	**115**	**111**	**107**	**103**
C2. Natural disaster	**101**	**101**	**98**	**95**	**92**	**90**	**87**	**85**	**82**	**80**	**77**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**101**	**96**	**94**	**89**	**86**	**84**	**82**	**79**	**76**	**74**	**71**
TOTAL public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
	PV of Debt-to-Revenue Ratio										
Baseline	677	637	619	598	583	569	554	538	523	508	492
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	677	712	700	681	669	659	648	637	626	615	604
B. Bound Tests											
B1. Real GDP growth	677	665	681	666	657	649	640	630	620	610	599
B2. Primary balance	677	666	681	658	640	624	607	590	572	555	537
B3. Exports	677	690	749	727	710	695	678	659	636	614	591
B4. Other flows 3/	677	652	650	628	612	598	583	566	549	532	515
B5. Depreciation	677	700	674	644	620	598	576	553	531	509	488
B6. Combination of B1-B5	677	645	661	640	624	610	595	579	563	547	530
C. Tailored Tests											
C1. Combined contingent liabilities	677	940	911	878	850	825	800	775	750	726	702
C2. Natural disaster	677	666	649	628	613	599	584	569	554	539	523
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	677	637	620	592	574	560	545	531	515	501	485
	Debt Service-to-Revenue Ratio										
Baseline	52	110	91	101	99	94	95	109	98	88	83
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	52	120	98	108	104	97	99	112	102	92	87
B. Bound Tests											
B1. Real GDP growth	52	114	98	110	108	104	107	124	113	104	99
B2. Primary balance	52	110	92	104	106	105	107	121	108	96	92
B3. Exports	52	110	93	107	105	100	101	117	110	100	94
B4. Other flows 3/	52	110	91	103	100	95	97	111	101	91	86
B5. Depreciation	52	113	99	111	107	100	102	117	107	97	91
B6. Combination of B1-B5	52	108	92	104	101	99	101	114	103	91	87
C. Tailored Tests											
C1. Combined contingent liabilities	52	110	102	108	155	149	148	168	125	120	122
C2. Natural disaster	52	112	92	103	104	99	101	115	102	93	88
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	52	110	98	111	101	84	95	108	98	88	83

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

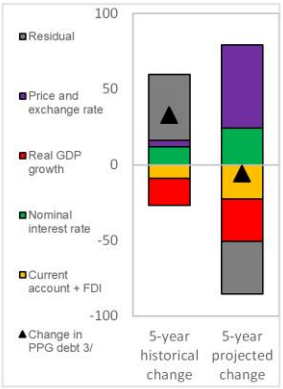
Figure 3. Drivers of Debt Dynamics—Baseline Scenario

External debt

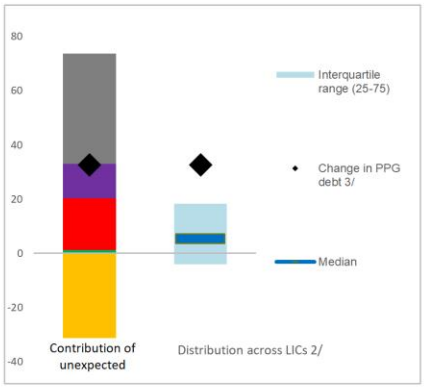
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

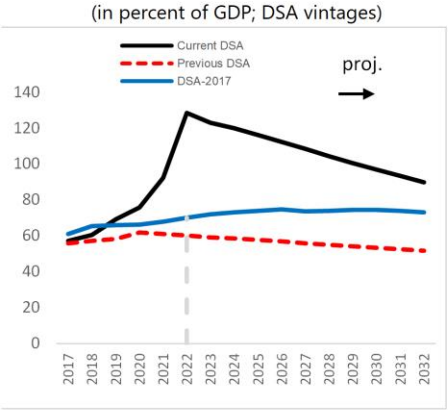


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

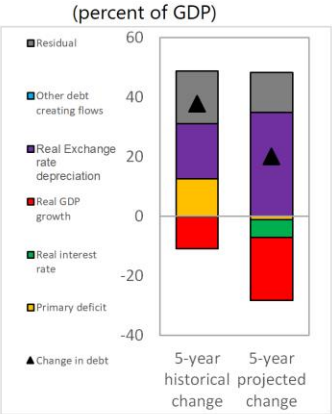


Public debt

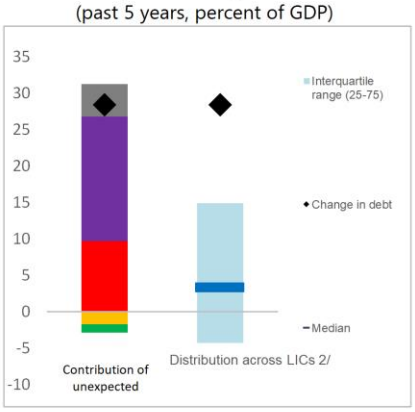
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



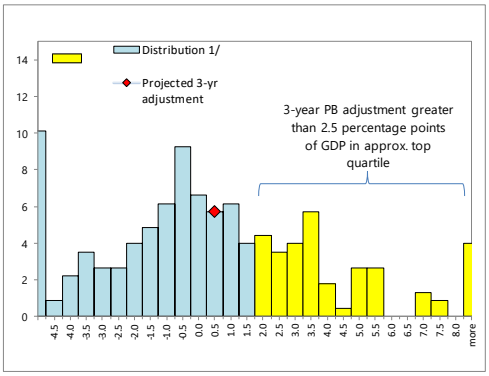
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

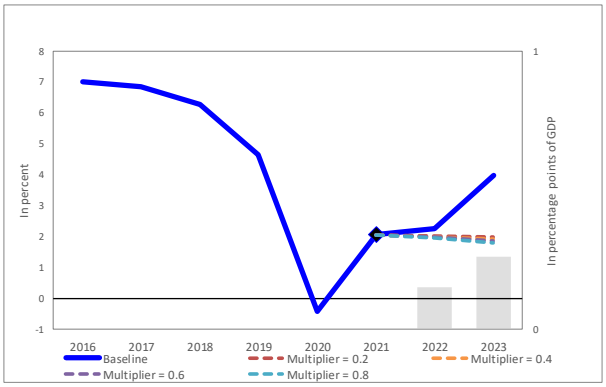
Figure 4. Realism Tools [1/]

3-Year Adjustment in Primary Balance (Percentage points of GDP)



1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



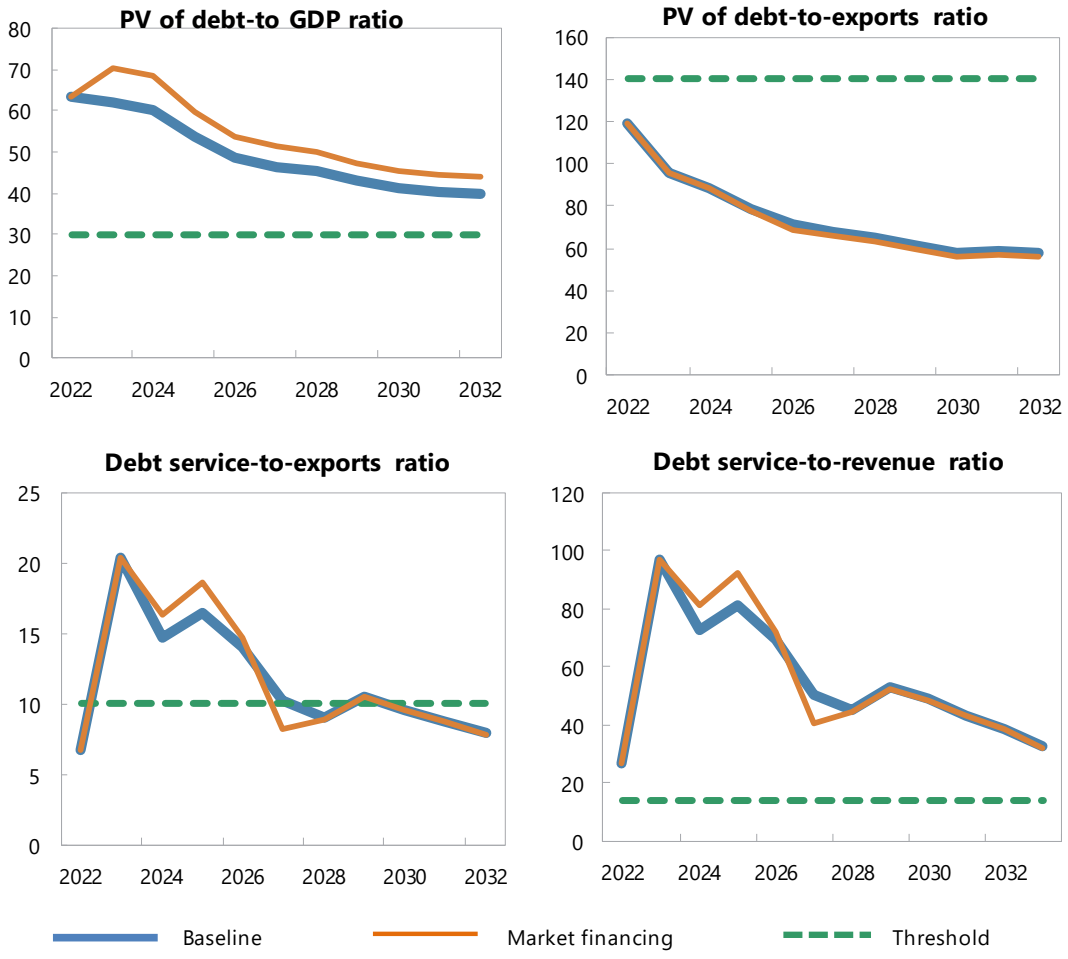
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

[1/] The data needed to conduct the investment growth tool is not available for the DSA exercise.

Figure 5. Market Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	16	n.a.
Breach of benchmark	Yes	n.a.
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



Sources: Country authorities; and staff estimates and projections.